

Paper Manager pack — RoboGen (Nov 18 · Data)

Paper: RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation
arXiv: <https://arxiv.org/abs/2311.01455> · [project](#) · [code](#) · **ICML 2024** (PMLR 235, Vienna)
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PDF: /workspace/papers/robogen-2311.01455.pdf (= /workspace/robogen-quiet/paper.pdf) · 48 pp · arXiv 2311.01455v3 (14 Jun 2024)
Session: Nov 18 · due 9:30 AM · Type: **Data**
Role weights: **ALL EMPTY** — still all five produced at balanced depth (~equal Empiricist / Reviewer / Visionary / Stakeholder / Archaeologist)
Kanta role: Produce **balanced** five-role pack (no PRIMARY seat); deepen motion + synthetic-data spines + Empiricist stage-isolated smokes
Themes: generative simulation · propose–generate–learn · synthetic skill demos opposite UMI/DROID/OXE real poles · BIT* MP primitives · long-horizon skill decomp
Sources: Summarizer (/workspace/papers/robogen-2311.01455-summary.md) + Senior Research (/workspace/robogen-quiet/DELIVERABLE.md) + paper_notes.md / env_outline/ — **Ablation Bot MISSING** (paper §4 + App B extract used)
Sibling links: /workspace/umi-quiet/ · /workspace/droid-quiet/ · /workspace/oxe-quiet/ · /workspace/diffusion-policy-quiet/ · /workspace/act-aloha-quiet/ · /workspace/dart-quiet/ · /workspace/pi05-quiet/ · /workspace/openpi-comet-smoke/

1. One-liner

****LLM proposes; SAC / BIT* / Adam learn — a generative sim factory that mints task–scene–reward–demo tuples so robots practice novel skills without real-world collection cost; complement**** (don't replace) UMI/DROID/OXE real poles for BEHAVIOR.

Contrast poles:

Pole	Who authors	Where	Product
RoboGen	LLM/VLM + physics sim	Generative sim (Objaverse / PartNet / soft / loco)	Endless synthetic skill demos + policies
UMI	Human + handheld gripper	Anywhere (cafe, fountain, home)	Action-rich real wrist demos
DROID	Robot teleop (Franka fleet)	564 real scenes / 52 buildings	Robot-in-wild real volume
OXE	Multi-lab agglomeration	22 robots / many labs	Cross-embodiment real mix
Behavior-100	Human designers	Hand-authored sim	Household ecology — RoboGen Table 1 foil

2. Summary (exec overview + key numbers)

RoboGen is a **generative robotic agent** that learns diverse skills via **generative simulation**. Foundation models do **not** emit torques — they **auto-generate tasks, scenes, and training supervisions**; physics + classical/learned optimizers solve. Self-guided **propose → generate → learn** cycle (Fig. 2):

- Task proposal** — GPT-4 seeded by robot + PartNet-Mobility / RLBench object (or 11 example tasks for loco/soft) → name, NL, assets, joints/links.
- Scene generation** — Objaverse top- $\backslash(k=\}10\backslash)$ Sentence-BERT + Gemini-Pro verify; GPT-4 sizes / joint inits / spatial; soft goals Midjourney → Zero-1-to-3 → DMTet.

3. **Training supervision** — GPT-4 decomposes; **selects** SAC RL / BIT* MP primitives / Adam traj-opt; writes rewards (EMD for soft).
4. **Skill learning** — sequential sub-tasks; each $\setminus N\{=8\}$ runs; best end-state → next init → demos.

Why Data session: Synthetic skill-demo factory opposite real poles — for novel articulated / soft / loco tasks where teleop is too costly. **Motion is PRIMARY:** BIT* primitives + algo-select (Fig. 5 RL-only collapse) + long-horizon decomp (avg **3.13** substeps).

Axis	Number / claim
Diversity Self-BLEU (↓)	0.284 (best vs Behavior-100 0.299 , RLBench 0.317 , GenSim 0.378)
SentBert / ViT / CLIP scene (↓)	0.165 / 0.193 / 0.762 — wins all Table 1 columns @ 106 tasks
Skill SR (human video)	0.774 on 69 tasks (manip 0.745/50 ; soft $\approx$ 0.886/7 ; loco $\approx$ 0.833/12)
Validity audit	19 / 155 fails (13 scene + 6 reward/decomp)
Fig. 5 algo	RL-only collapses on 12 articulated; MP+RL selection critical
Long-horizon structure	Avg 3.13 substeps (up to 8–10); $\sim$ 1.5 RL + $\sim$ 1.63 MP
Wall-clock (App.)	Plan $\leq$ 10 min /task; RL subgoal $\sim$ 2–3 h ; typical 4–5 h (8×2.5 GHz CPUs)
SAC knobs	MLP [256] ³ , lr 3e–4 , 1M /sub, horizon 100 /fs 2 , 6D EE
MP knobs	BIT* (OMPL); pre-contact 0.03 m along normal
Soft / loco	Adam 300 /lr 0.05 /EMD; CEM horizon 150 /fs 4
GPT temps	Proposal 0.8–1.0 ; other stages 0–0.3

Must-cite tattoo: diversity **0.284/0.165/0.193/0.762** · SR **0.774** (69) · fails **19/155** · substeps **3.13** · Fig. 5 **RL-only collapse** · ****LLM proposes; SAC/BIT*/Adam learn****.

Do NOT claim: zero human (prompts + ICL + human video judge); sim-to-real solved (§5 open); **106 = infinite** (evaluated finite snapshot; pipeline can mint more).

3. Keywords

RoboGen; generative simulation; propose-generate-learn; GPT-4; Gemini-Pro; Objaverse; PartNet-Mobility; SAC; BIT*; OMPL; motion planning primitives; Adam traj-opt; CEM; soft-body EMD; long-horizon subtasks; skill library; N=8 handoff; GenSim; Behavior-100; Genesis; infinite synthetic data; sim-to-real gap; BEHAVIOR synthetic factory; UMI/DROID/OXE complement

4. Deep dive — Motion / skill-library / planner-vs-RL (PRIORITY)

Callout: RoboGen's long-horizon competence is **decomp + choose MP or RL or traj-opt per sub-skill**, not a single end-to-end LLM policy. ****BIT* motion planning is first-class****, not garnish — Fig. 5 proves it.

Algorithm menu (§3.3–3.4, App A.3) — CLEAR

Algo	When	Mechanism
BIT* MP primitives	Approach / grasp / release; collision-free path	Sample surface → align gripper to normal → plan to pre-contact (0.03 m) → push to contact; OMPL BIT*
SAC RL	Contact-rich / continuous / non-pose-parameterizable (knob, loco-ish contact)	6D EE (Δ xyz + Δ axis-angle); privileged state; MLP [256] ³ ; lr 3e–4 ; 1M steps/sub
Adam traj-opt	Fine soft-body shaping	300 grad steps; lr 0.05 ; EMD to target particles; horizon 150/200
CEM loco	Legged skills (paper finds more stable than RL here)	GT dynamics; horizon 150 , fs 4 ; joint-angle actions

Long-horizon / skill-library structure (Fig. 3, Fig. 6, App B.1)

Knob	Paper	Empiricist use
Decomposition	GPT-4 → shorter-horizon sub-tasks	Chain planner primitives; log stage SR
Retries	Each sub-task $\backslash(N\{=8\})$ runs	Best-of handoff — smoke may cut to N=2–4
State glue	Highest-reward end-state → next init	Sequential skill atoms
Avg structure	3.13 total; ~1.5 RL + ~1.63 MP	Hybrid is the product
Long tail	Up to 8–10 substeps	Smoke $\leq 3-4$; defer 8–10

"Skill library" status: Paper emits **demos + skills** (inventory / demo stream) — packaged reusable API library = **TENTATIVE**. CLEAR as proto skill-library atoms with algo tags.

Fig. 5 — CRITICAL ablation (motion FLAG)

On 12 articulated tasks: **RL-only completely fails for most tasks**; full **algorithm-selection** (MP primitives ± RL) restores success.
Moral: Primitives encode grasp/approach structure that RL must otherwise discover — report action space / prim availability carefully (Reviewer confound).
Empiricist P0: force RL-only vs allow MP on **one** short articulated open/close stub (not full 12-task suite).

Motion exam bite

"RoboGen = LLM decomp + ****choose BIT* or SAC or Adam per sub-skill; avg 3.13 substeps (~1.5 RL + 1.63 MP); $\backslash(N\{=8\})$ best-state handoff**; Fig. 5 RL-only collapse** — MP is PRIMARY."

5. Deep dive — Synthetic data factory (PRIMARY DATA CONTRIBUTION)

FLAG: RoboGen's scientific product is an **endless stream of skill demonstrations** (policies + trajectories) tied to generated tasks/scenes — i.e. **synthetic data + learned skills**, not a static offline dump. Session type = Data; opposite UMI/DROID/OXE **real** poles.

Propose–generate–learn components (Fig. 2) — CLEAR

Stage	Backend	Output
A. Task proposal	GPT-4 T 0.8–1.0; object-based (PartNet/RLBench) or example-based (11 tasks)	Name, NL, assets, joints/links
B. Scene gen	Objaverse $\backslash(k\{=10\})$ + Gemini-Pro caption → GPT-4 verify; LLM sizes; collision push (App A.2)	Valid sim scene + soft text → 3D goals
C. Supervision	GPT-4 T 0–0.3; 3 ICL reward examples	Sub-tasks + algo choice + reward code
D. Skill learn	SAC / BIT* / Adam / CEM on Genesis (internal)	Policies + demos; sequential handoff

Design doctrine (CLEAR §3): Extract **semantics / affordances / common sense** from FMs — **not joint torques / contact dynamics**. Backend LLM/VLM modules are swappable.

Diversity claim — Table 1 (106 RoboGen tasks)

Metric (↓ better)	RoboGen	Behavior-100	RLBench	MetaWorld	ManiSkill2	GenSim
# Tasks	106	100	106	50	20	70
Self-BLEU	0.284	0.299	0.317	0.322	0.674	0.378
SentBert emb sim	0.165	0.210	0.200	0.263	0.194	0.288
Scene ViT emb sim	0.193	0.389	0.375	0.517	0.332	0.717
Scene CLIP emb sim	0.762	0.833	0.864	0.867	0.828	0.932

CLEAR: Wins all ↓ columns at matched ~100-task scale. GenSim gap = GenSim's tabletop pick-place + small Ravens pool vs RoboGen articulated/loco/soft + Objaverse.
Do not equate 106 evaluated with "infinite" — pipeline can mint endlessly; evaluation is a finite snapshot.

Validity + verification (Fig. 4, App B.3)

Check	Result
Scene human fail	13 / 155 (missing functionality; joint≠semantic open/closed; precision pairs)
Reward/decomp fail	6 / 155 (undefined vars; inverted fold/unfold; continuous rhythmic motions)
Total	19 / 155 ≈ 12%
Fig. 4 size verify	w/o size → drastic BLIP-2 drop (dominant)
Fig. 4 object verify	w/o object → lower mean + higher variance

Synthetic vs real poles (session genealogy)

Axis	RoboGen	UMI / DROID / OXE
Authoring cost	LLM prompts + ICL (+ audit)	Human / robot teleop logistics
Novelty coverage	High for novel artic/soft/loco once prompts exist	Bound by where humans/robots go
Visual/tactile realism	Sim — gap admitted \$5	Real — gold for deploy
Product form	Endless demo stream	Finite collected corpora
BEHAVIOR use	Practice ground / coverage	Challenge teleop / VLA priors

Downstream consumers (siblings): Diffusion Policy / ACT / $\pi_{0.5}$ / OpenPI FT on exported demos; DART recovery noise optional; OXE mix discipline (source=robogen); UMI orthogonal at collection UX — combine only at Visionary recipe layer.
Do not claim: zero human involvement; sim-to-real solved; generative demos are deployment-ready without gap proxy.

6. Role-ready sections (all five — balanced)

6.1 Empiricist — BALANCED (stage-isolated generative-sim smokes)

Seat charge: Course smokes that stress proposal validity · object/size verify · planner-vs-RL · gap proxy, ranked by GPU/RAM/TIME/API honesty. Prefer stage-isolated stubs on a public sim stand-in — not full paper-scale propose → learn / Genesis soft fleet / 1M×106.

Claim under measurement

Can an LLM-orchestrated propose → verify → hybrid-solve → export loop produce **directionally** valid scenes and learnable short skills on public-sim stubs, with Fig. 4/5 morals reproducible without claiming paper **0.774** / Table 1 parity?

Hypotheses (falsifiable — from DELIVERABLE)

1. **H-algo**: On contact-rich articulated stubs, **MP-primitive ± short RL** $\gg$ **RL-only** SR (Fig. 5 moral).
2. **H-size**: Dropping **size verification** hurts asset-text alignment more than dropping object verify alone (Fig. 4).
3. **H-object**: Object verification reduces retrieval variance / false accepts vs language-only match.
4. **H-proposal**: Stage-A/B stub yields $\geq 70\%$ schema-valid, collision-resolved scenes (paper scene fail $\sim 13/155 \approx 8\%$ — smoke bar looser).
5. **H-decomp**: LLM sub-task orderings are mostly executable with planner primitives on short chains (Fig. 3).
6. **H-reward**: Static reward gens have non-trivial syntax/semantic fail rate; execute+repair (App B.3) lifts validity.
7. **H-coverage**: Generative task text @ matched N has **lower** Self-BLEU / emb-sim than fixed benchmark labels (Table 1 moral at small N).
8. **H-gap**: Policy trained on privileged/clean synth **drops** under appearance/camera DR — quantifies synthetic $\leftrightarrow$ real **proxy** gap paper leaves open.
9. **H-downstream**: Tiny RoboGen-like demos as aug to DP/ACT/ π_0 help novel-task SR **or** hurt if gap large — sibling Partial.

Falsifiers: RL-only $\approx$ hybrid; size/object verify noop; proposal chaos $\gg$ paper fail; decomp unusable; rewards always valid; generative $\neq$ more diverse at small N; DR no drop; synthetic aug always helps.

Prioritized ablation matrix (paper extract — Ablation Bot MISSING)

Pri	Experiment	Paper hook	Smoke	Est. GPU-h / cost
P0	Planner+RL hybrid vs RL-only	Fig. 5 collapse	Y	2–12 or CPU MP minutes
P0	Object verify $\pm$ size verify	Fig. 4 BLIP-2	Y	0–1 + VLM API \$
P0	Task/scene proposal stub	Fig. 2 A–B	Y	API \$ / CPU
P0	Sim-to-real gap proxy	\$5 limitation	Y (proxy)	1–8
P1	Tiny synthetic skill train	App B.2 / Tables 5–7	Y	2–8
P1	Decomposition quality	Fig. 3	Partial	API + CPU
P1	Novel-task coverage card	Table 1	Partial	API + CPU
P1	Reward validity audit	App B.3	Y	API + CPU
P2	Retrieval vs text-to-3D	Design §3	Partial/N	High
P2	Demos $\rightarrow$ DP/ACT/ π_0 FT	Downstream siblings	Partial	8–24
Defer	Full 106 propose $\rightarrow$ learn	Full system	N	$\gg 100$
Defer	Genesis soft / 12-loco CEM	Tables 6–7	N	Needs Genesis/alt

Default course smoke order:

proposal_stub $\rightarrow$ verify_ablation (full / w/o object / w/o size) $\rightarrow$ planner_vs_rl (hybrid vs RL-lite) $\rightarrow$ gap_proxy $\rightarrow$ optional skill_tiny_synth / diversity_card.

Exact reproduce stack (public stand-in — EDIT)

```
# Official (verify current; soft/Genesis may be incomplete)
git clone https://github.com/Genesis-Embodied-AI/RoboGen
# Project: https://robogen-ai.github.io/
# Paper Genesis was internal at publication – course smoke uses PUBLIC stand-in

# Course scout env (quiet outlines – preferred)
export ROBOPEN_ROOT=$HOME/robogen-smoke
export OPENAI_API_KEY=...          # EDIT if live LLM stages
export GEMINI_API_KEY=...          # EDIT if verify stages
cd /workspace/robogen-quiet
bash env_outline/setup_mamba.sh
# mamba activate robogen-smoke
mkdir -p logs runs
# EDIT partition/account + sim stand-in (ManiSkill2 / Isaac / PyBullet / MuJoCo) in slurm_*.sh
# Outlines only – do NOT sbatch from quiet box
# Pin model IDs + sim name → logs/pins.txt
# NEVER imply Genesis bit-parity
```

Run cards (RC0–RC5)

RC0 — `proposal_stub` (API/CPU gate)

Field	Spec
Goal	Prove LLM → valid task+scene JSON
Input	Prompt template + robot/object seed (or mock LLM)
Steps	Propose N=10–20 → schema validate → optional collision push → log
Pass	≥70% schema-valid + collision-resolved; wall 10–40 min + API
Fail	Chaos >> paper 8% scene-fail moral; missing required fields
Script	<code>env_outline/slurm_proposal_stub.sh</code>

RC1 — `verify_ablation` (Fig. 4)

Field	Spec
Goal	Size (± object) verify ↑ alignment
Setup	Freeze 5–7 retrieved meshes; toggle object VLM verify + size LLM rescale
Metrics	BLIP-2 / CLIP-score / human checklist
Expected	Size verify dominates; object verify ↓ variance
Budget	0–1 GPU-h + VLM API; no skill train
Script	<code>env_outline/slurm_verify_ablation.sh</code>

RC2 — `planner_vs_rl` (Fig. 5) — MOTION P0

Field	Spec
Goal	Hybrid >> RL-only on one articulated stub
Setup	Open drawer / close microwave stand-in; BIT*/IK approach vs SAC-lite ≤50–100k (cut paper 1M)
Metrics	SR / return; #solved substeps
Expected	RL-only fails contact-heavy; hybrid lifts
Budget	2–12 GPU-h or CPU MP-only minutes
Script	<code>env_outline/slurm_planner_vs_rl.sh</code>

RC3 — skill_tiny_synth (P1)

Field	Spec
Goal	Stub rewards + SAC/BC learn one short skill
Setup	Planner demos → BC or SAC ≤100k; automated proxy (joint thresh / contact)
Pass	Reward curve ↑; proxy SR logged — do not claim 0.774
Budget	2–8 GPU-h
Script	env_outline/slurm_skill_tiny_synth.sh

RC4 — gap_proxy (P0/P1)

Field	Spec
Goal	Quantify synthetic ↔ real proxy gap
Setup	Train on privileged/synth → eval under texture/camera DR
Metrics	ΔSR; feature distance
Expected	Drop under DR — honest gap card before "infinite data" rhetoric
Budget	1–8 GPU-h
Script	env_outline/slurm_gap_proxy.sh

RC5 — diversity_card (P1 Partial)

Field	Spec
Goal	Table 1 moral at small N
Setup	N≈20–50 generated vs same-N subsample of RLBench/MetaWorld labels
Metrics	Self-BLEU; SentBert/CLIP emb-sim
Expected	Generative lower (better) Self-BLEU / emb-sim directionally
Budget	API + CPU <1 h; do not claim exact Table 1 scalars
Script	env_outline/slurm_diversity_card.sh

Paper hyperparams to pin (App A.3)

Knob	Paper	Smoke cut
SAC	lr 3e−4, MLP [256] ³ , 1M/sub, H=100/fs=2	≤50–100k
BIT*	pre-contact 0.03 m	keep
GPT T	proposal 0.8–1.0 / other 0–0.3	pin model IDs
Soft Adam	300 / lr 0.05 / EMD	Defer (needs Genesis/alt)
CEM loco	H=150 / fs=4	Defer
Retrieval	⋈ _{k={10}}	freeze tiny shard

Exact skill SR slices (exam copy — human video judge)

Slice	n	Mean SR
Articulated/rigid manip (Table 5)	50	0.745
Soft-body (Table 6)	7	≈0.886
Locomotion (Table 7)	12	≈0.833
All benchmarked	69	0.774

Failure modes cheat-sheet

Symptom	Likely cause	Next action
RL-only $\approx$ hybrid	Primitives unavailable / easy pose-only task	Switch to contact-rich artic stub
Size verify noop	Assets already correctly scaled	Use deliberately wrong-size baselines
Proposal chaos	Temp too high / bad ICL	Drop T; add schema repair loop
Claimed 0.774 from stub	Overclaim	Report stub proxy SR only
CUDA false under GPU reservation	Misconfigured env	Fail closed (run_checklist §2)
"Reproduced RoboGen"	Full-loop rhetoric on Partial	Document stage costs; refuse slogan

What not to invent / claim

- **Do not** claim zero human (prompts + ICL + human video SR).
- **Do not** claim sim-to-real solved (§5 open — gap proxy only).
- **Do not** equate **106** evaluated with infinite capacity without noting snapshot.
- **Do not** amortize paper **0.774** onto Partial stubs.
- **Do not** imply Genesis bit-parity from public stand-in.
- **Do not** invent Ablation-Bot authority — file **MISSING**; cite paper extract.

Empiricist exam bite

"RoboGen = propose $\rightarrow$ verify $\rightarrow$ hybrid-solve $\rightarrow$ export; Fig. 5 **MP critical** (RL-only collapses); Fig. 4 **size verify** dominates; SR **0.774/69** human-judged; fails **19/155**; smoke RC0 $\rightarrow$ RC2 directional — never claim zero-human / sim-to-real / 106=infinite."

6.2 Scientific Reviewer — BALANCED

Claims under review

1. Generative simulation can match/surpass **human-crafted** benchmark diversity.
2. Extracting semantics (not actions) from FMs + physics learning is the right division of labor.
3. Automatic **algo selection** (esp. MP primitives) is necessary for articulated success.
4. Pipeline yields **valid** scenes and supervisions often enough to learn skills (~77% run success).

Strengths

1. **Modality discipline** — LLMs for semantics/affordances/tasks, **not** torques — first-class vs "LLM policy" papers.
2. Table 1 diversity win at matched scale is clean and multi-metric (text + image).
3. Fig. 5 RL-only ablation is a strong causal argument for hybrid MP+RL.
4. Failure appendix (App B.3 / Table 8) is unusually honest — raises credibility.
5. Scope beyond GenSim (articulated affordances, soft-body, loco) is substantive.
6. Full-stack integration (task + scene + reward + algo select + learn) > piece-wise concurrent ideas.
7. Design is backend-swappable in principle (GPT/Gemini/Midjourney named but not permanently wired).

Weaknesses / threats

1. Success rates are **human-judged videos**, not automated sensors — rater variance unreported.
2. 106-task diversity eval vs "endless" rhetoric: evaluated finite snapshot.
3. Backend GPT-4 / Gemini / Midjourney are **closed, moving targets** — reproducibility fragile.
4. **No quantitative sim-to-real** transfer experiment (§5 admits gap).
5. **Genesis opacity** — internal differentiable sim undermines soft-body reproducibility; treat as existence proof.
6. Fig. 5 confound: hybrid wins partly because **primitives encode structure** RL must discover — report action space carefully.
7. Diversity metrics (Self-BLEU / emb-sim) can reward verbosity/noise; pair with human "useful skill" ratings.

- 8. GenSim comparison fairness — different task class / asset pool; cite as **coverage**, not pure algo superiority.
 - 9. Verification stack dependency (Gemini + GPT-4 + BLIP-2) — backend churn moves Fig. 4.
 - 10. Cost externalities: "minimal human" ≠ "minimal compute/\$" (endless API + CPU).
 - 11. No experiment training a **generalist** (OXE/DP/VLA) on RoboGen streams — downstream assumed, not shown.
 - 12. Ablation Bot catalog **MISSING** this session — factorial LLM/backend/step ablations absent from paper too.
- Verdict:** Strong **systems + paradigm** paper for ICML. Core story (FM for semantics → sim for physics; hybrid learners) holds. Do **not** upgrade to "solved infinite robot data" or "BEHAVIOR data problem closed."
- Reviewer exam bite:** Demand Fig. 5 + Fig. 4 before "RoboGen just works"; cite **0.774** as human-video on 69, not automatic; punish "106 = infinite" and sim-to-real solved claims.

6.3 Visionary — BALANCED (BEHAVIOR synthetic / generative-sim recipe; no Discord/Drive)

BEHAVIOR connection:
 BEHAVIOR / Behavior-100 / Behavior-1K embody **sim-first household data hunger** — historically human-authored. RoboGen Table 1 literally ranks against **Behavior-100** and wins diversity, while citing Behavior-1K as the scalability pain generative simulation attacks. For **2026 BEHAVIOR**-class settings, RoboGen is a **design prior** for automating task+scene+demo minting — **complement** to UMI handheld / DROID robot-in-wild / OXE real mix, not a replace.

BEHAVIOR need	RoboGen-shaped response
Task coverage across kitchens/rooms	Auto task proposal seeded on PartNet appliances + distractors
Long-horizon activities	Sub-task decomp + sequential skill chaining as proto skill library
Contact-rich articulated skills	Keep MP primitives + RL menu (Fig. 5 lesson)
Soft / deformable household	Traj-opt + EMD + text → 3D goals
Mix with real / VLA priors	RoboGen demos as sim column in OXE-style soup; don't drown real wrist demos
Verification	Add VLM success critics before demos enter the mix (paper's own open problem)

CLEAR: Design prior for BEHAVIOR-like automation.
TENTATIVE: Official BEHAVIOR'26 baseline names RoboGen (organizer checklist not verified).

2026 BEHAVIOR generative sim / synthetic data recipe (10 bets)

- Stage-isolated factory:** Ship **propose** → **verify** → **plan/RL** → **export demo** with pinned models — steal the **pipeline**, not Genesis lock-in.
- Verification is the product:** Size + object + joint-state semantic checks (App B.3 taxonomy) before any SAC budget.
- Planner-first, RL sparsely:** Match Fig. 5 — spend RL only on contact/dynamics subgoals; export demos for VLA FT.
- Mix card with real:** Generative demos as labeled shard beside BEHAVIOR teleop / OXE priors — Empiricist leave-synthetic-out ablation.
- Gap proxy mandatory:** Every synthetic shard publishes **DR / appearance ΔSR** (H-gap) before "infinite data" rhetoric.
- Novel-task eval:** Hold out LLM-proposed tasks never seen in teleop; report coverage vs Self-BLEU vanity.
- Long-horizon glue:** LLM decomp as **System-2 stages** compatible with BEHAVIOR stage votes / commit_s (siblings ACT/DP).
- Do not retrain SAC-MLP** as the 2026 leaderboard policy — **export demos** → $\pi_{0.5}$ / **N1.7**.
- Sibling stack:** DROID scene-turnover + RoboGen task invent + DART recovery noise + OXE mix discipline.
- One narrative line:** Hand suites taught **benchmarking**; OXE/DROID taught **real diversity**; RoboGen taught **automate the sim curriculum**; 2026 should **schedule generative practice** into the data mix with **verify + gap cards**.

Follow-up research

1. Closed-loop **VLM verifiers** for scene / joint-state / skill success (kill 19/155 modes).
2. Eureka-style **reward refinement** with env feedback inside the loop.
3. Export demos into **RLDS / Open-X** schema (`source=robogen`).
4. **Skill library** packaging: name, language, pre/post, algo tag, demo blobs — hierarchical VLA atoms.
5. Domain rand + rendering for **measurable** sim-to-real on held-out real appliances.
6. Open backends + frozen versions for longitudinal reproducibility.
7. Mobile-manip / multi-robot seeding beyond Franka + quadruped.
8. Active querying: propose tasks that maximize coverage gaps vs BEHAVIOR/OXE mix.

New applications

- Rapid appliance skill farms (microwave, dishwasher, safe) before real teleop.
- Soft-body kitchen prep curricula as stress tests for diff-sim + VLA FT.
- Quadruped loco skill zoos as synthetic pretrain for parkour.
- Overnight curriculum generators for courses / challenges.
- Synthetic hard-negatives for VLA eval without safety risk.

Visionary one-liner: RoboGen turns BEHAVIOR's **authoring bottleneck** into a **queryable factory** — infinite sim demos where UMI/DROID cannot afford to go — then hands hierarchical VLAs a growing **skill library** if verification and schema export catch up.

Visionary exam bite: "For BEHAVIOR'26, steal RoboGen's **propose** → **verify** → **hybrid-solve** → **export**; keep MP-first (Fig. 5); publish gap proxies; race on $\pi_{0.5}/N1.7$ — don't retrain SAC-MLP as the entry; **complement** UMI/DROID/OXE, don't replace."

(No Discord / Drive checklist content.)

6.4 Stakeholder — BALANCED (produced despite empty seat)

Product one-liner: An LLM-orchestrated **sim factory** that mints task–scene–reward–demo tuples so robots practice novel skills without real-world collection cost.

Who cares

- Labs / companies needing **many novel household skills** without teleop armies.
- Sim-platform teams (Genesis / Isaac / MuJoCo) wanting an **auto task+scene+reward** layer.
- BEHAVIOR-class challenge teams hungry for **coverage** before scarce gold demos.
- **Non-consumers:** need certified **real** demos only; no sim budget; require verified sim-to-real today.

Buy / build decision

1. Bottleneck = **authoring tasks/rewards/scenes** in sim → adopt RoboGen-style propose–generate–learn.
2. Bottleneck = **real visual/tactile diversity** → still fund UMI/DROID/OXE; use RoboGen as **sim prior / skill bootstrap**.
3. Bottleneck = **tabletop pick-place only** → GenSim-class may suffice; RoboGen's value is articulated/soft/loco + algo selection.
4. Always budget **human audit** (~12% fail on 155) + verification — do not ship unattended endless minting into training mixes.

Risks: GPT-4/Gemini API cost & drift; Genesis availability; silent reward inversions; sim-to-real tax; verification not automated at scale; "minimal human" masking high compute/\$.

Stakeholder exam bite: "RoboGen buys endless **sim** skill coverage; UMI/DROID/OXE buy **real** diversity — pick the pole, then budget the verify + gap day; never ship unattended minting."

6.5 Archaeologist — BALANCED (produced despite empty seat)

Priors RoboGen absorbs (CLEAR)

- Sim skill learning: PyBullet / MuJoCo / SoftGym / FluidLab / DiffSkill / RoboNinja lineage.
- Human sim benchmarks: **RLBench**, **Meta-World**, **ManiSkill2**, **Behavior** / **Behavior-1K**.
- LLM-for-robotics: Code-as-Policies, Inner Monologue, VoxPoser, Language-to-Rewards, **Eureka**, SayCan.
- Concurrent foils: **GenSim** (LLM writes manip code; tabletop); **Gen2Sim** (Katara et al.).
- Assets: PartNet-Mobility / SAPIEN, Objaverse, Zero-1-to-3, Midjourney.
- Planners/learners: SAC, BIT*/OMPL, CEM, Adam traj-opt.
- Paradigm: **Generative Simulation** — Xian et al. 2023a (2305.10455).

Genealogy (CLEAR framing)

```
Human-authored sim benchmarks (RLBench / Behavior-100)
  + FM semantic knowledge (GPT-4 / Gemini)
  + Generative Simulation paradigm (Xian et al. 2023a)
    → RoboGen propose-generate-learn
      → synthetic skill demos / policies
        || UMI / DROID / OXE real poles
          ↓
        BEHAVIOR / VLA mixes that need
        verify + gap cards on synthetic shards
```

Contrast poles (session lineage):

OXE / DROID / UMI = **real data** → generalist VLAs;

RoboGen / GenSim = **generative sim data** → skill coverage where real collection is too costly;

BEHAVIOR = **sim-first household challenge** that still needs heroic human authoring **or** RoboGen-like automation.

Descendants / ecosystem

- Public **RoboGen** PyBullet repo + robogen-ai.github.io — **CLEAR**.
- **Genesis** engine (announced; release timing **TENTATIVE**).
- Conceptual influence on later LLM-reward + generative-task systems — do not over-claim unique descent without citation checks.

Genealogy one-liner: Behavior-style human sim authoring || GenSim code-gen → **RoboGen full-stack generative simulation** (scenes + rewards + MP/RL/traj-opt) → infinite synthetic skill stream **opposite** UMI/DROID real poles.

Archaeologist exam bite: "RoboGen = GPT-4 proposes tasks/scenes/rewards; physics + SAC/BIT*/Adam learn; diversity beats Behavior-100 on Table 1; ~77% skill-run success; **MP ablation matters**; complement real poles."

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export ROBOGEN_WORK_ROOT=$HOME/robogen-smoke
export OPENAI_API_KEY=...    # EDIT
export GEMINI_API_KEY=...    # EDIT
cd /workspace/robogen-quiet
bash env_outline/setup_mamba.sh
# mamba activate robogen-smoke
mkdir -p logs runs
# EDIT: public sim stand-in + partition/account in slurm_*.sh
# Pin model IDs + sim name → logs/pins.txt
# Outlines only – do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	robogen-smoke sketch (Torch + SB3/SAC + sentence-transformers/CLIP; no Genesis pin)
env_outline/setup_mamba.sh	Create/update env; EGL/OSMesa; API key placeholders
env_outline/run_checklist.md	GPU / TIME / API honesty preflight
env_outline/slurm_proposal_stub.sh	P0 task/scene proposal N=20
env_outline/slurm_verify_ablation.sh	P0 object/size verify (Fig. 4)
env_outline/slurm_planner_vs_rl.sh	P0 hybrid vs RL-lite (Fig. 5)
env_outline/slurm_skill_tiny_synth.sh	P1 tiny synthetic skill / BC
env_outline/slurm_gap_proxy.sh	P0/P1 appearance/DR gap proxy
env_outline/slurm_diversity_card.sh	P1 Self-BLEU / emb-sim small-N
env_outline/README.md	Cluster usage notes

GPU / RAM / TIME budget honesty

Assumptions: Empiricist path = **stage-isolated stubs** on a **public** sim, **not** paper Genesis + 1M SAC/sub × dozens of tasks. Paper: planning ~≤10 min CPU; each RL subgoal ~2–3 h (8 threads @ 2.5 GHz); typical task 4–5 h; 32-core ≈ 8 parallel jobs — **mostly CPU sim**. LLM/VLM stages dominate **API \$ + latency**, not VRAM. SAC MLP is small GPU.

Workload	GPU VRAM (peak)	Host RAM	Wall time (order)	Smoke
Proposal stub N=20	0	8 GB	10–40 min + API	Y
Object/size verify @ 5–7	0–8 GB or API	16 GB	30–90 min	Y
MP-primitive only 1 task	0–4 GB	16 GB	minutes–1 h	Y
SAC-lite ≤50–100k 1 sub	4–12 GB	16–32 GB	1–4 h	Y
Paper SAC 1M/sub × ~1.5 RL	4–12 GB	32 GB	~2–3 h × 1.5/task	Partial (cut)
Diversity Self-BLEU N=50	0–8 GB emb	16 GB	<1 h	Y
Gap proxy (BC + DR)	8–16 GB	32 GB	2–8 h	Y
Soft-body traj-opt suite	depends	32 GB+	hours×7	N
Full 106 propose → learn	mixed	large	days–weeks	N
Downstream DP/ACT/ π_0 FT	12–48 GB	64 GB	8–24 h	Partial

Honesty rule: log (stage, model_id, T, n_tasks, env_steps, algo, verify_flags, wall_s, api_tokens/\$, metric). If #SBATCH --gres=gpu:N but CUDA false → **fail**. Never amortize paper 0.774 onto a Partial stub.

Course envelope: target ≤ ~20–25 GPU-h + capped API \$ for P0 matrix (RC0–RC2 + gap). Full paper-scale = ≫100 GPU-h + \$ — do not promise.

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
Endless diverse skill stream	Proposal stub valid varied tasks @ N=20	106-task Table 1 regen
Diversity ≥ human suites	Directional Self-BLEU / emb-sim on small N	Exact Table 1 scalars @ N=106
Scene validity via verify	Fig. 4 moral: size (± object) ↑ alignment	Exact BLIP-2 bars
Training supervisions induce skills	1 stub skill learns under gen reward or planner chain	Fig. 3 four LH videos
Algo select > RL-only	Hybrid > RL-lite on one articulated card	Fig. 5 12-task suite
~0.774 skill SR	Automated proxy; report stub SR honestly	Human-video 69-task average
Minimal human supervision	Count checklist minutes on smoke	"Zero human" claim
Sim-to-real gap exists	Gap proxy shows ΔSR under DR	Real Franka deploy
Soft / loco coverage	Note-only	Tables 6–7 reproduction

Pass: ≥2 P0 cards documented (verify and/or algo and/or proposal) with stage cost logs.

Fail / overclaim: "reproduced RoboGen / matched 0.774 / beat Behavior-100 diversity" from stubs without matched

protocol.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Empiricist	balanced · ~14	Hypotheses, RC0–RC5, reproduce cmds, GPU/API honesty, ablation matrix
Reviewer	balanced · ~14	Fig. 4/5 bite; human-video SR; Genesis opacity; verdict paradigm Accept
Visionary	balanced · ~14	BEHAVIOR propose → verify → export recipe; no Discord/Drive
Stakeholder	balanced · ~14	Buy/build vs real poles; audit budget; risks
Archaeologist	balanced · ~14	Genealogy Behavior GenSim → RoboGen UMI/DROID/OXE

Spine (paste into Slide Maker) — 5 bullets:

1. ****LLM proposes; SAC / BIT* / Adam learn**** — generative sim factory mints task–scene–reward–demo tuples.
2. **Must-cite numbers:** diversity **0.284/0.165/0.193/0.762** (Table 1 @106); SR **0.774/69**; fails **19/155**; avg **3.13** substeps.
3. **Motion PRIMARY:** BIT* MP primitives + algo-select; Fig. 5 **RL-only collapse**; $\backslash(N\{=\}8\backslash)$ best-state handoff; ~1.5 RL + 1.63 MP.
4. **Synthetic data:** endless skill-demo stream opposite **UMI/DROID/OXE** real poles; Fig. 4 size/object verify critical; complement don't replace.
5. **BEHAVIOR'26 + honesty:** propose → verify → hybrid-solve → export; gap proxies mandatory; never claim zero-human / sim-to-real solved / 106=infinite; Empiricist RC0–RC2 directional smokes.

GitHub title sketch: [ECE 605] - RoboGen Making Balanced Five

Full Slide Maker brief: /workspace/robogen-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/robogen-quiet/KANTA_PACK.md
PDF	/workspace/robogen-quiet/KANTA_PACK.pdf
Slide brief	/workspace/robogen-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/robogen-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/robogen-2311.01455-summary.md
Ablations	MISSING — use DELIVERABLE §2 / paper_notes extract
Paper notes	/workspace/robogen-quiet/paper_notes.md
Env outlines	/workspace/robogen-quiet/env_outline/
Paper PDF	/workspace/papers/robogen-2311.01455.pdf
Page figs	/workspace/papers/robogen-figs/page-*.png

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 Paper Manager — Nov 18 Data · ALL seats empty · all five balanced · Ablation Bot absent · no Discord/Drive.